Neap

VCE Specialist Mathematics Units 3&4

Question and Answer Booklet

2025 Trial Examination 2

Reading time: 15 minutes

Writing time: 2 hours

Student's Name: _____

Teacher's Name: _____

Approved materials

- Protractors, set squares and aids for curve sketching
- One bound reference
- One approved CAS calculator or CAS software, and one scientific calculator

Materials supplied

- Question and Answer Booklet of 24 pages
- Formula Sheet
- Multiple-Choice Answer Sheet

Instructions

- Write **your name** and your **teacher's name** in the spaces above on this page.
- Follow the instructions on your Multiple-Choice Answer Sheet.
- At the end of the examination, place your Multiple-Choice Answer Sheet inside the front cover of this booklet.

Students are **not** permitted to bring mobile phones and/or any unauthorised electronic devices into the examination room.

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Section B (6 questions, 60 marks) 10–24

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Section A – Multiple-choice questions

Instructions

- Answer **all** questions in pencil on your Multiple-Choice Answer Sheet.
- Choose the response that is **correct** for the question.
- A correct answer scores 1; an incorrect answer scores 0.
- Marks will **not** be deducted for incorrect answers.
- No marks will be given if more than one answer is completed for any question.
- Unless otherwise indicated, the diagrams in this booklet are **not** drawn to scale.
- Take the **acceleration due to gravity** to have magnitude g m s^{-2}, where $g = 9.8$.

Question 1

Consider the following statement.

If m is odd, then $m^2 + 3m - 4$ is even.

What is the contrapositive of this statement?

A. If $m^2 + 3m - 4$ is even, then m is odd.

B. If $m^2 + 3m - 4$ is odd, then m is even.

C. If m is even, then $m^2 + 3m - 4$ is even.

D. If m is odd, then $m^2 + 3m - 4$ is odd.

Question 2

The graph of $f(x) = \dfrac{x^2 + px + 1}{x + 1}$ has no turning points when

A. $p < 2$

B. $p < -2$

C. $p > 2$

D. $p > -2$

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Question 3

Let $z = 100 + 100i$.

The values of n such that $z^{n+1} \in R$ are

A. $4k, k \in Z$

B. $8k + 1, k \in Z$

C. $8k - 1, k \in Z$

D. $4k - 1, k \in Z$

Question 4

If $z = a + 2bi$ and $w = -1 - 2i$, where $a > b > 0$, then $\text{Arg}\left(\frac{\bar{z}w}{2i}\right)$ is equal to

A. $\pi - \tan^{-1}\left(\frac{a+4b}{2(a-b)}\right)$

B. $\tan^{-1}\left(\frac{a+4b}{-2(a-b)}\right)$

C. $-\tan^{-1}\left(\frac{a+4b}{2(a-b)}\right)$

D. $\tan^{-1}\left(\frac{a+4b}{2(a+b)}\right)$

Question 5

Consider the differential equation $\frac{dy}{dx} = e^{x-y}\sin(x)$, where $y\left(\frac{\pi}{2}\right) = \frac{\pi}{2}$.

Euler's method is applied to the equation using a step size of 0.1.

The difference between the actual value of y and the value of y_5 is closest to

A. 0.0004

B. 0.004

C. 0.04

D. 0.4

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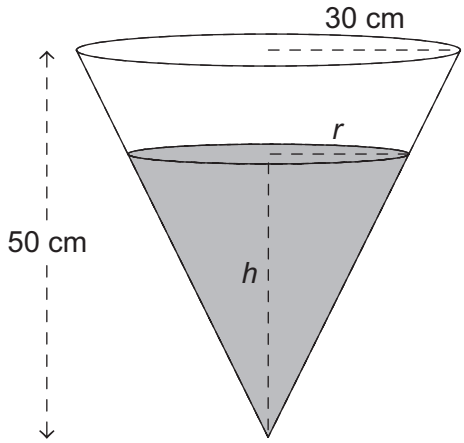
Question 6

An inverted cone is shown in the diagram below.



The cone is initially three-quarters full of water. The water begins leaking out the bottom of the cone at a rate of $\frac{k\sqrt{h}}{2}$ mL min^{-1}. As the water leaks, the radius of the surface is monitored.

When the depth of the water is 20 cm, the rate of change of the radius, in cm min^{-1}, is given by

A. $\frac{3k\sqrt{5}}{2000\pi}$

B. $-\frac{3k\sqrt{5}}{2000\pi}$

C. $\frac{k\sqrt{5}}{240\pi}$

D. $-\frac{k\sqrt{5}}{240\pi}$

Question 7

Given that $\cos(x) = -\frac{1}{b}$, where $x \in [-\pi, 0]$ and $b \in R^+$, the value of cosec(2x) is

A. $\frac{2\sqrt{b^2-1}}{b^2}$

B. $-\frac{2\sqrt{b^2-1}}{b^2}$

C. $\frac{b^2}{2\sqrt{b^2-1}}$

D. $-\frac{b^2}{2\sqrt{b^2-1}}$

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Question 8

Using integration by parts, $\int x^{n^2} \log_e(nx)\,dx$, where $n \in N$, can be expressed as

A. $\dfrac{x^{n^2+1}}{n^2+1}\left(\log_e(nx) - \dfrac{1}{n}\right) + c$

B. $\left(\dfrac{x\log_e(nx)}{n^2+1} - \dfrac{x}{(n^2+1)^2}\right)x^n + c$

C. $\dfrac{x^{n^2+1}}{n^2+1}\left(\log_e(nx) - \dfrac{1}{n^2+1}\right) + c$

D. $x^{n^2}(x\log_e(nx) - x) - \int n^2x^2\log_e(nx) - x\ dx + c$

Question 9

A curve is defined by the parametric equations $x = 2\cos\left(\dfrac{t}{3}\right) + 2t$ and $y = -\sin^2\left(\dfrac{t}{3}\right) + t^2$.

The length of the curve for $t \in [0, \pi]$ is closest to

A. 6.62

B. 11.14

C. 41.07

D. 44.79

Question 10

From sea level, a projectile is fired vertically down and a steel ball is dropped into the ocean.

The actions occur simultaneously.

The initial velocity of the projectile is 30 m s^{-1}, and it experiences a deceleration of a m s^{-2}, where $a = -\dfrac{v}{20}$. The steel ball is dropped from rest, and its acceleration is constant at 3 m s^{-2}. Both objects travel in a straight line.

After t seconds, the depth of the projectile is five times the depth of the ball.

The distance between the two objects after t seconds is closest to

A. 68 m

B. 72 m

C. 80 m

D. 96 m

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Question 11

The velocity of a particle moving along an east–west line, relative to the fixed point *O*, at time *t* seconds is given by $v = 2\cos(t^2)\sin(t^2)$.

The initial position of the particle is 2 m east of *O*.

After two seconds, the displacement of the particle from its initial position is closest to

A. –1.55 m

B. 0.45 m

C. 1.55 m

D. 2 m

Question 12

Consider the vectors $\underset{\sim}{a} = 2\underset{\sim}{i} - \underset{\sim}{j} + 3\underset{\sim}{k}$, $\underset{\sim}{b} = -\underset{\sim}{i} + 4\underset{\sim}{j} + n\underset{\sim}{k}$ and $\underset{\sim}{c} = 2\underset{\sim}{i} + \underset{\sim}{j} + m\underset{\sim}{k}$, where $m, n \in R^+$.

Given that the vectors are linearly dependent and $|\underset{\sim}{b}| = |\underset{\sim}{c}|$, the values of *m* and *n* are

A. $m = \dfrac{4\sqrt{37} + 63}{11}$, $n = \dfrac{7\sqrt{37} + 36}{11}$

B. $m = \dfrac{-(4\sqrt{37} - 63)}{11}$, $n = \dfrac{-(7\sqrt{37} - 36)}{11}$

C. $m = -\dfrac{4}{9}$, $n = \dfrac{7}{9}$

D. $m = 7$, $n = 19$

Question 13

Let *P* be the point (3, –4, –1), which lies on the plane *x*–*y*–*z*, and *X* be the point on the *x*-axis that is closest to *P*.

Which of the following could be the vector equation of a line that goes through *P* and *X*?

A. $\underset{\sim}{r}(t) = t(3\underset{\sim}{i} - 4\underset{\sim}{j} - \underset{\sim}{k})$

B. $\underset{\sim}{r}(t) = 3\underset{\sim}{i} - 4\underset{\sim}{j} - \underset{\sim}{k} + t(-\underset{\sim}{j} - 4\underset{\sim}{k})$

C. $\underset{\sim}{r}(t) = 3\underset{\sim}{i} + t(-\underset{\sim}{j} - 4\underset{\sim}{k})$

D. $\underset{\sim}{r}(t) = 3\underset{\sim}{i} - 4\underset{\sim}{j} - \underset{\sim}{k} + t(4\underset{\sim}{j} + \underset{\sim}{k})$

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Question 14

The plane Π is defined by the equation $2x - 3y + z = 18$. Let X be the point of intersection between Π and line L, which has the cartesian equation $\frac{1-x}{2} = y - 3 = \frac{z}{2}$.

Point $A(1, 3, 0)$ also lies on line L. The line segment AX forms the hypotenuse of a right-angled triangle. One other side of the triangle lies completely within Π.

The area of the triangle is closest to

A. 35

B. 40

C. 45

D. 50

Question 15

Given that $\underset{\sim}{v} = m\underset{\sim}{a} + n\underset{\sim}{b}$ and $\underset{\sim}{w} = -2\underset{\sim}{a} + 3\underset{\sim}{b}$, the vector cross product $\underset{\sim}{v} \times \underset{\sim}{w}$ is

A. $\underset{\sim}{0}$

B. $(3m + 2n)(\underset{\sim}{a} \times \underset{\sim}{b})$

C. $(3m + 2n)(\underset{\sim}{b} \times \underset{\sim}{a})$

D. 0

Question 16

A tank initially contains 1000 L of water that has 2.5 kg of salt dissolved in it.

A solution containing 85 grams of salt per litre is poured into the tank at a rate of 10 litres per minute. The solution is stirred continuously. The tank is also being drained at a rate of R litres per minute.

Let x be the total mass, in kilograms, of salt in the tank.

Which of the following differential equations describes the rate of change of x after t minutes?

A. $\frac{dx}{dt} = 850 - \frac{Rx}{1010 - R}$

B. $\frac{dx}{dt} = 8.5 - \frac{Rx}{1000 - R}$

C. $\frac{dx}{dt} = 850 - \frac{Rx}{1000 - R}$

D. $\frac{dx}{dt} = 8.5 - \frac{Rx}{1010 - R}$

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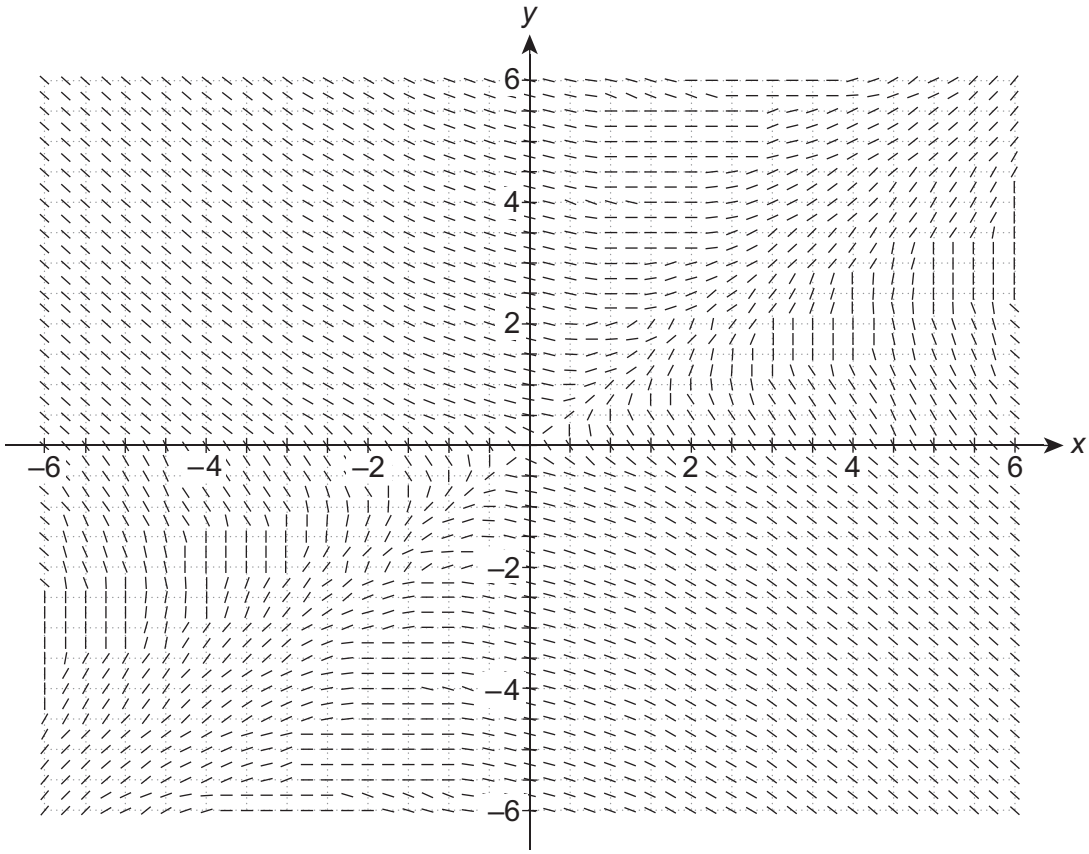
Question 17

A slope field is shown below.



The slope field is best described by

A. $\dfrac{dy}{dx} = \dfrac{2x - y}{2y - x}$

B. $\dfrac{dy}{dx} = \dfrac{2y - x}{2x - y}$

C. $\dfrac{dy}{dx} = 2y - x$

D. $\dfrac{dy}{dx} = 2x - y$

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Question 18

Consider the functions T and B.

$$T(x) = \frac{4x^2 + 3}{2x^2 + 2}, x \geq 0$$

$$B(x) = \frac{x^2 + 300}{200}, x \geq 0$$

The region bounded by T and B is rotated about the x-axis to obtain a solid of revolution.

The volume of this solid is closest to

A. 2.63

B. 8.16

C. 9.25

D. 29.07

Question 19

A factory manufactures table legs that should be 99 cm long. A quality inspector carries out a one-sided statistical test to determine the actual length of the table legs and concludes that, at the 5% level of significance, the table legs are shorter than 99 cm.

Given that the sample mean is 97.9 cm and the standard deviation is 5.5 cm, what is the minimum sample size?

A. 67

B. 68

C. 69

D. 70

Question 20

A farmer shears their flock of sheep once a year. Initially, the mean mass of wool the sheep produce is 4.5 kg, with a standard deviation of 160 g.

The farmer purchases a new type of sheep feed that is designed to increase wool mass. After the new food is introduced, a sample of 25 sheep is taken and the mean mass of the wool produced is 4.57 kg.

The farmer should conclude that, at the 1% level of significance, the food

A. does not affect wool mass as the p value is above 0.01.

B. does not affect wool mass as the p value is below 0.01.

C. increases wool mass as the p value is above 0.01.

D. increases wool mass as the p value is below 0.01.

End of Section A

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Section B

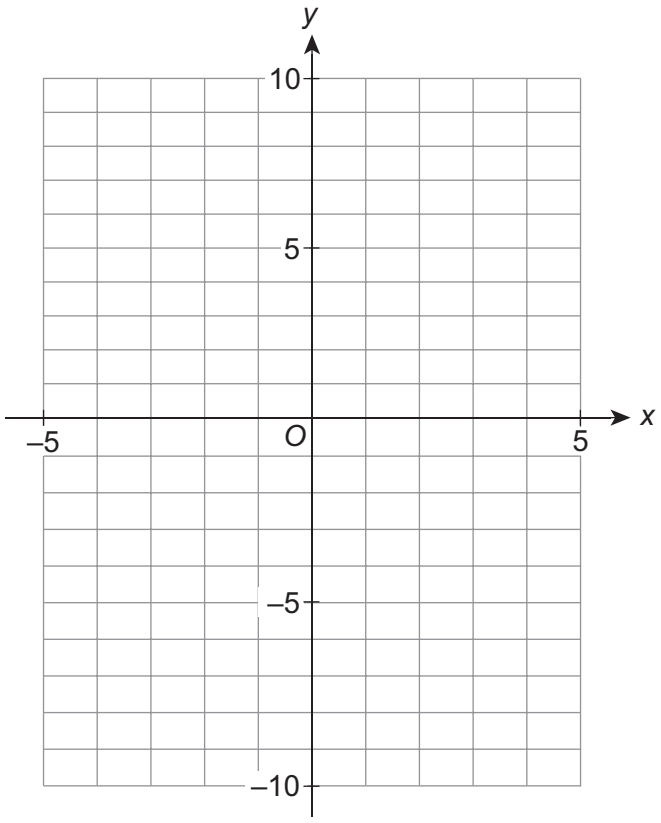
Instructions

- Answer **all** questions in the spaces provided.
- Write your responses in English.
- Unless otherwise specified, an **exact** answer is required to a question.
- In questions where more than one mark is available, appropriate working **must** be shown.
- Unless otherwise indicated, the diagrams in this booklet are **not** drawn to scale.
- Take the **acceleration due to gravity** to have magnitude g m s^{-2}, where $g = 9.8$.

Question 1 (10 marks)

Consider the function f with rule $f(x) = \dfrac{x^5 + x^2 - 2}{x^2 - 1}$.

a. Sketch the graph of $y = f(x)$ on the set of axes below. Label the vertical and oblique asymptotes with their equations and label the stationary points with their coordinates, correct to one decimal place. 3 marks



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b. The region bounded by the graph of f and the lines $x = \frac{3}{2}$ and $x = 2$ is rotated about the x-axis to produce a solid of revolution.

i. Write down a definite integral for the curved surface area of the solid in terms of x. 2 marks

ii. Find the total surface area of the solid, including the caps at each end of the curved surface. Give your answer correct to one decimal place. 1 mark

c. **i.** Find the length of the curve between $x = \frac{3}{2}$ and $x = 2$. Give your answer correct to two decimal places. 1 mark

ii. The length of the curve between $x = \frac{3}{2}$ and $x = 2$ is equal to the length of the curve between $x = 2$ and $x = q$.
Find the value of q, where $q > 2$. Give your answer correct to two decimal places.
1 mark

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d. The region bounded by the graph of f and the lines $x = 0$ and $x = \frac{1}{2}$ is rotated about the x-axis to produce a solid of revolution.

Find the volume of the solid. Give your answer correct to two decimal places. 2 marks

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Question 2 (11 marks)

When sketched on an Argand diagram, the relation $|z-2+i|=3$, where $z \in C$, intersects with the imaginary axis twice.

a. Find the distance between the two points of intersection. 3 marks

b. The imaginary axis divides the graph of the relation into two segments.

Find the area of the smaller segment. Give your answer correct to two decimal places. 2 marks

The ray described by $\text{Arg}(z+1)=k$ passes through the positive point of intersection between the imaginary axis and the relation.

c. Find the value of k. Give your answer in radians, correct to two decimal places. 2 marks

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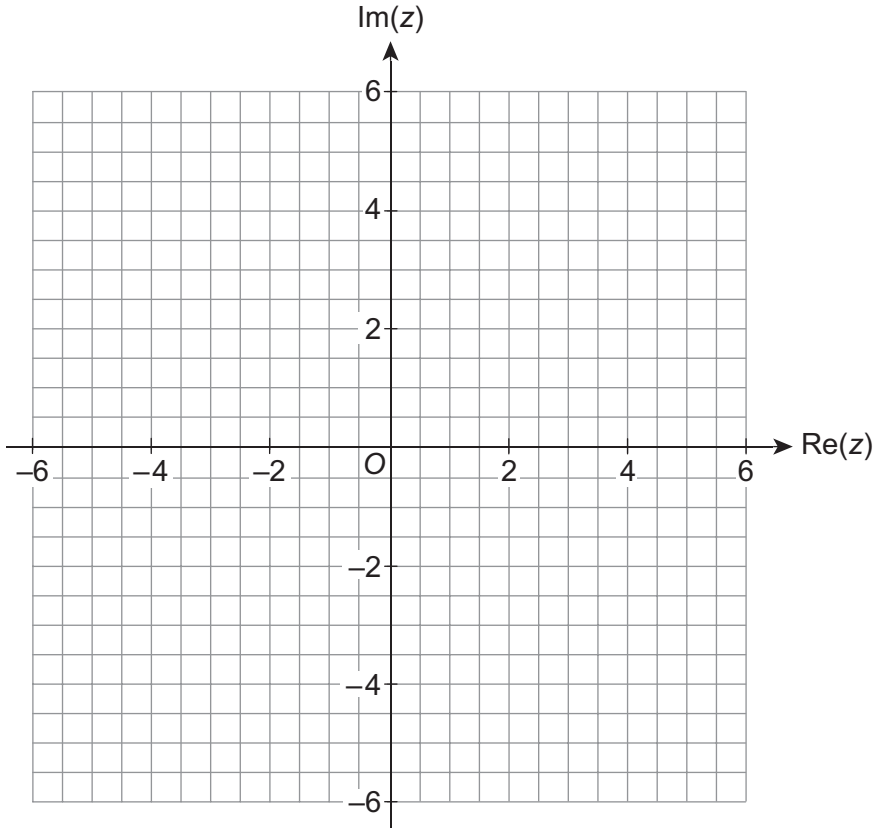
d. Sketch the relation and the ray on the Argand diagram below. Label any axis intercepts with their coordinates. 2 marks



e. The smallest circle that can pass through the points $-1 + 0i$ and $0 + \left(1 + \sqrt{5}\right)i$ has the Cartesian equation $(x + h)^2 + (y + k)^2 = m$.

Find the values of h, k and m. 2 marks

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Question 3 (8 marks)

The price of a particular company's stock, AAMT, has a rate of change of $\frac{dP}{dt} = 2t^2\cos(t)$.

Let P be the price of AAMT t months after it entered the stock market, where P is measured in cents and t is measured in radians.

a. Use integration by parts to show that $P(t) = 2t^2\sin(t) + 4t\cos(t) - 4\sin(t) + c$. 2 marks

b. When it first entered the stock market, AAMT was traded at a price of 60 cents.

i. Find the price of AAMT, in cents, after it has been in the stock market for five months. Give your answer correct to one decimal place. 1 mark

ii. Find the rate of change of the price of AAMT after it has been in the stock market for six months. Give your answer in cents per month, correct to one decimal place. 1 mark

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The price of another stock, BBMT, has a rate of change of $\frac{dP_B}{dt} = 0.3P_B\left(1 - \frac{P_B}{350}\right)$, where P_B is the price of the stock, in cents, after t months. When BBMT entered the stock market, its price was 10 cents. Assume that BBMT entered the stock market at the same time as AAMT.

c. During the period $0 < t < 8$, determine the number of months where the price of BBMT is greater than the price of AAMT. Give your answer correct to two decimal places. 3 marks

d. The price of a third stock, CCMT, has a rate of change of $\frac{dP_C}{dt} = kP_C\left(1 - \frac{P_C}{350}\right)$, where P_C is the price of the stock, in cents, after t months. When CCMT entered the stock market, its price was 15 cents. Assume that CCMT entered the stock market at the same time as AAMT and BBMT.

Find the maximum value of k such that the price of CCMT is never greater than the price of AAMT during the period $0 < t < 8$. Give your answer correct to three decimal places. 1 mark

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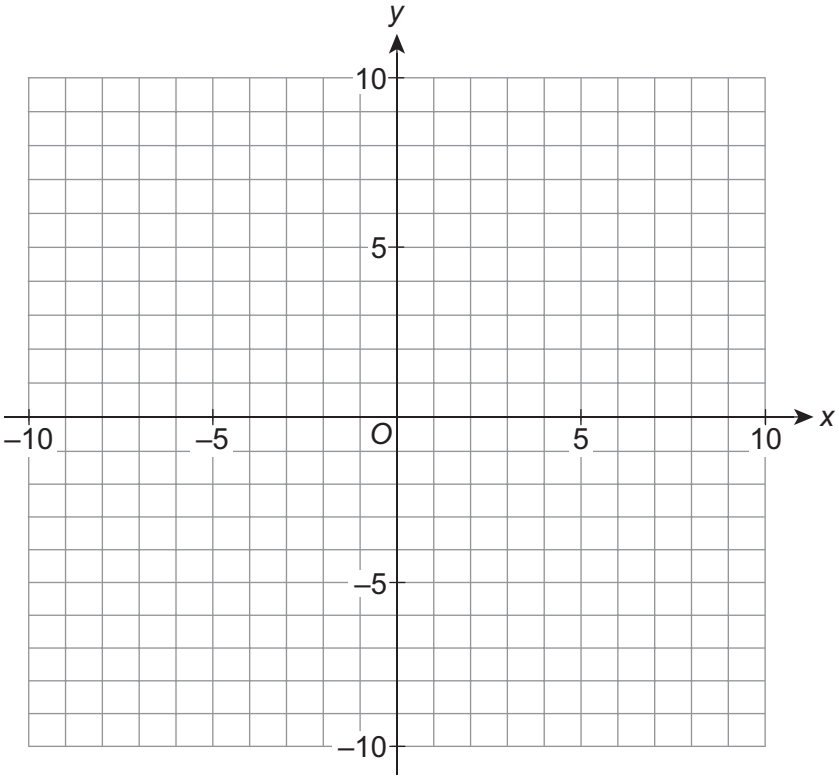
Question 4 (10 marks)

The path of a remote-controlled car along a racing track, relative to the fixed point *O,* is given by $\underset{\sim}{r}_C = 10\sin(t)\underset{\sim}{i} + 2\sin(2t)\underset{\sim}{j}$, where $0 \le t \le 2\pi$.

The components of the car's displacement are measured in metres, and time *t* is measured in seconds.

a. Find the Cartesian equation of the path of the car. 2 marks

b. Sketch the path of the car on the axes below. Label the axis intercepts with their coordinates and show the direction of motion. 2 marks



c. **i.** Find an expression for the speed of the car in terms of t. 1 mark

ii. State the maximum speed of the car. 1 mark

iii. Find the minimum speed of the car and the times at which the car reaches the minimum speed. 2 marks

d. A drone is hovering directly above the point $Q\left(-5\sqrt{2}, -2\right)$ on the car's path. At time $t = 0$, the drone drops a large rock from its position in the air. The rock falls straight down and lands on the track at the same time that the car drives through point Q. The rock is subject only to gravitational acceleration.

Determine the height of the drone above the ground. Give your answer in metres, correct to two decimal places. 2 marks

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Question 5 (12 marks)

The line L_1 has the vector equation $\underset{\sim}{r} = 3\underset{\sim}{i} + 4\underset{\sim}{j} + t\left(-\underset{\sim}{i} - \underset{\sim}{j} + 2\underset{\sim}{k}\right)$.

a. Find the acute angle between L_1 and the plane defined by the x- and y-axes. Give your answer in degrees, correct to two decimal places. 2 marks

b. Find the shortest distance between L_1 and the z-axis. 2 marks

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c. Consider the points $A(0, 0, 3)$, $B(3, 0, 1)$ and $C(9, 3, 3)$.

Find the acute angle at which the plane containing A, B and C intersects with the plane defined by the x- and z-axes. Give your answer in degrees, correct to one decimal place. 3 marks

d. Find the Cartesian equation of a plane that is parallel to L_1 and passes through the origin. 2 marks

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The plane $x + y + z = 0$ is parallel to L_1.

e. Find the shortest distance between L_1 and the plane $x + y + z = 0$. 2 marks

f. Find the cartesian equation of a plane that L_1 lies within. 1 mark

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Question 6 (9 marks)

A factory produces a detergent called Mauve, which is sold in bottles. According to the label, each bottle of Mauve contains 700 mL of detergent. The volumes of the detergent in each bottle of Mauve are normally distributed, with a standard deviation of 10.5 mL.

A consumer protection agency is investigating the company. Under consumer law, the company can be fined if, on average, the bottles contain a smaller volume of detergent than advertised.

A representative takes a sample of 20 bottles and conducts a one-sided statistical test at the 5% level of significance. They obtain a mean volume of 695.7 mL.

a. Write down the null hypothesis, H_0, and the alternative hypothesis, H_1, for the test. 1 mark

b. **i.** Find the p value for the test, correct to three decimal places. 1 mark

ii. Will the company be fined? Give a reason for your response. 1 mark

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The factory produces a different detergent called Cyan, which is also sold in 700 mL bottles. Cyan is a mixture of two ingredients, compound A and compound B. The volumes of compound A in each bottle are normally distributed, with a mean of 400 mL and a standard deviation of 3.5 mL. The volumes of compound B are also normally distributed, with a mean of 300 mL and a standard deviation of 4 mL.

c. Find the probability that a bottle of Cyan has a volume greater than 705 mL. Give your answer correct to three decimal places. 2 marks

d. Cyan is sold to supermarkets in boxes of 24 bottles.

Find the probability that the mean volume of the bottles in one box is between 698 mL and 702 mL. Give your answer correct to three decimal places. 2 marks

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e. A customer buys one bottle of Mauve and one bottle of Cyan. Assume that the mean volume of detergent in a bottle of Mauve is 700 mL.

What is the probability that the bottle of Mauve contains at least 5 mL more detergent than the bottle of Cyan? Give your answer correct to two decimal places. 2 marks

End of examination questions

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